Improving Detection and Segmentation Performance of Prohibited Items Detection Task Models

Abdulaziz Alqahtani

Supervisor: Dr. Qasim Umer

Abstract

Prohibited-item detection in X-ray baggage screening is essential for security checkpoints, yet it remains difficult to automate reliably because X-ray imagery exhibits weak texture, low contrast, and strong semi-transparent superposition, where objects overlap and visually mix. These conditions amplify two deployment-critical failure modes: false negatives (missed threats, especially when items are concealed or partially visible) and false positives (background-triggered alarms on benign, cluttered configurations). This proposal targets improved reliability for prohibited-item detection and instance segmentation by combining (i) a Systematic Literature Review (SLR) and (ii) a controlled experimental study on a realistic benchmark. First, the SLR will map datasets, supervision types (boxes/masks), overlap-aware modeling, and generalization issues, and will distill recurring limitations that motivate a focused research target. Second, we will establish reproducible baselines on PIDray, leveraging its difficulty splits (easy/hard/hidden) to evaluate robustness under deliberate concealment, using an efficiency-oriented CNN detector (YOLO11s) and a real-time transformer-based detector (RT-DETRv2) models. Finally, we will conduct diagnostic error analysis (false positives/false negatives, split-wise and per-class trends, and overlap-induced localization ambiguities) to identify where failures concentrate and to motivate practical, background-aware directions for reducing clutter-driven errors while preserving real-time constraints for screening deployments.

Introduction

Security screening (e.g. X-ray baggage scanning in airports, public spaces, etc.) is critical in maintaining a safe and flowing environment by detecting prohibited items and ensuring public safety in a reasonable amount of time [1]. Traditional baggage screening by human operations lead error prone, particulary under high passenger transport rate. Missed threats may lead to catastrophic security breaches, while items that have been flagged as threats slow down the flow of checkpoints. Therefore, there is strong impetus to develop computer vision models that can robustly detect and segment illicit items in X-ray images with high sensitivity and precision. However, X-ray images is mainly differs from photographic images in ways that pose new challenges. The degree to which X-rays are absorbed by various materials varies, producing color or intensity indications that reveal composition or density [2]. Real-world X-ray images often suffer from overlapping objects and cluttered backgrounds, which makes the automatic detection of prohibited items extremely challenging due to visual ambiguity [3]. Compared to everyday photographs, the difference in appearance is significant; a gun or knife in an X-ray image looks markedly different and more indistinct in shape than it does in a color photograph [4].

In the context of prohibited item detecion, the field has two main vision tasks: object detection, which uses a class label and a bounding box to locate each danger item, and instance segmentation, which additionally specifies the exact mask of every object. Both tasks face failure modes under real world X-ray conditions. The biggest concern is the possibility of false negatives, which occur when a danger item that is obscured or visually subtle is missed. False positives, in which harmless items (such as gadgets or litter) are mistakenly identified as threats and cause unnecessary alarm, are equally troublesome. The overlapping nature of X-ray images makes both problems worse: a prohibited item hidden behind other objects may not produce a distinct enough signal to be detected, and a bag's contents mixed up can fool a detector into activating on harmless configurations [1]. For segmentation models, heavy occlusion leads to boundary instability; the model may fails to merge near objects or chop a single object to fragments when their X-ray silhouettes mix [4]. Therefore, hte key to improve performance in this domain is to achieve robustness against clutter and overlap.

Methodologically, prohibited-item detection in X-ray screening largely inherits the broader evolution of object detection: two-stage region-proposal pipelines (Fast R-CNN [5] and Faster R-CNN [6]), one-stage dense detectors designed for speed (e.g., YOLO [7] and SSD [8]), and more recent transformer-based formulations (e.g., DETR [9] and hierarchical vision transformers such as Swin [10]). Instance segmentation typically builds on these detectors by adding a mask-prediction branch, with Mask R-CNN remaining a canonical baseline [11]. When transferred to baggage X-ray imagery, however, superposition and weak texture can amplify background confusion and blur object boundaries, so X-ray-specific studies often augment generic backbones with domain modules, such as de-occlusion attention and lateral inhibition—to suppress overlap-induced feature mixing and refine edges [15, 16]. Still, many approaches remain modular refinements of general architectures, leaving robust performance on deliberately concealed cases and practical efficiency constraints as persistent requirements for real screening deployments [13].

Problem Statement

A key difficulty is that heavy clutter and transparent overlap introduce feature mixing between foreground threats and background structures, producing two dominant failure modes: false negatives, where true prohibited items are suppressed or predicted as background, especially when items are concealed or partially visible, and false positives, where confusing background regions or benign object configurations trigger alarms. These errors become more severe in deliberately challenging cases, such as "hard" and "hidden" settings in PIDray evaluation, where small or occluded threats are difficult to isolate from surrounding baggage contents.

Although prior work reports strong benchmark performance using advanced detection pipelines, a practical gap remains in understanding and reducing background-triggered misclassification under clutter when using modern detectors within limited computational budgets. In particular, it is not yet clear how lightweight CNN-based detectors and the transformer era real-time detectors differ in their robustness to overlap-induced ambiguity and background confusion when trained and evaluated under the same protocol. Therefore, improving prohibited-item detection for X-ray screening requires methods and diagnostics that explicitly target background confusion and overlap-driven ambiguity while maintaining realistic efficiency constraints for deployment.

Related Work

Dataset Context and Annotation Scope

X-ray prohibited-item benchmarks have grown from early, small-scale datasets to larger and more task-diverse corpora that better reflect real baggage screening, where superposition, clutter, and weak texture degrade both detection and mask boundaries. Key datasets used in this area are summarized in Table 1. Mery *et al.* introduced GDXray as an early public dataset resource [12], and Miao *et al.* released SIXray at much larger scale to study prohibited-item discovery under overlap and extreme class imbalance [14]. Zhang *et al.* expanded supervision with PIDray by providing large-scale annotations that support both detection and instance segmentation, together with difficulty splits (easy/hard/hidden) to evaluate concealed threats [13].

Occlusion-focused benchmarks further emphasize overlap-heavy screening. Wei *et al.* introduced an occluded prohibited-item detection benchmark (OPIXray) and studied occlusion-aware modeling in that setting [15], while Zhao *et al.* proposed CLCXray (cutters and liquid containers) and validated a label-aware mechanism on both CLCXray and OPIXray [18]. Tao *et al.* proposed HiXray for large-scale airport screening and evaluated detection performance in conjunction with OPIXray [16]. For segmentation, Ma *et al.* released PIXray as a public instance-level dataset [17], later extended by Ren *et al.* via PIDray-A/PIXray-A to support occlusion-aware mask modeling with additional occlusion annotations [35]. Beyond single-view supervision, Tao *et al.* introduced an X-ray few-shot benchmark to study rare threats [26] and later the dual-view LDXray dataset for multi-view detection under heavy occlusion [19], while Zhang *et al.* curated a balanced SIXray+PIDray subset for controlled imbalance studies [41].

Dataset	Year	Images	Classes	Task
GDXray [12]	2015	19,407	3	Detection
PIDray [13]	2023	124,486	12	Detection + Segmentation
SIXray [14]	2019	1,059,231	6	Classification + weak/partial localization
OPIXray [15]	2020	8,885	5	Detection
HiXray [16]	2021	45,364	8	Detection
PIXray [17]	2022	5,046	15	Detection + Segmentation
CLCXray [18]	2022	9,565	12	Detection
LDXray [19]	2024	146,997	12	Dual-view Detection

Table 1: X-ray Prohibited-Item Detection Datasets

Core Challenges in X-ray Prohibited Item Understanding

Prohibited-item understanding in X-ray imagery is challenging because objects are semi-transparent and often overlap, causing feature mixing and unclear boundaries. X-ray scans also tend to exhibit weak texture, low contrast, and cluttered layouts, which makes small items easier to miss and increases the likelihood of confusing background structures for threats [12, 14, 16, 17].

Prior work addresses these issues through overlap-aware modeling, feature enhancement, and data-centric strategies. Occlusion-focused designs are explored in the context of OPIXray/CLCXray-style evaluation [15, 18], and suppression-style mechanisms have been studied to reduce distracting responses in cluttered baggage [16]. Other approaches emphasize extracting more reliable cues when appearance is weak, including feature-aware detection designs [28], material/physics-inspired cues [29, 36], and augmentation strategies that explicitly target background clutter [44]. For segmentation, the core difficulty remains mask ambiguity under severe overlap; public instance-level datasets and occlusion-extended variants highlight the need for methods that improve boundary quality under superposition [17, 35].

Detection and Segmentation Approaches

Many approaches build on standard detection backbones and introduce X-ray-oriented enhancements for overlap and small objects. One common direction refines one-stage detectors with attention or multi-scale feature fusion, including attention-enhanced YOLO variants [20, 21, 22, 34, 40] and lightweight models tailored to constrained screening settings [30]. Attention-based prohibited-item detection has also been explored beyond these specific YOLO refinements [24, 25], and general deep-learning detection studies remain useful as practical reference points [27].

Transformer-style and hybrid architectures provide an alternative path when overlap induces strong feature coupling. AO-DETR and MMCL represent recent efforts that explicitly target anti-overlap detection within DETR-like frameworks [43, 39], while comparative work on hybrid CNN–transformer configurations highlights robustness trade-offs across datasets and scanner conditions [33]. For open-world operation, open-vocabulary detection via CLIP adaptation has been investigated to handle emerging categories without requiring the same closed-set

labeling assumptions [31]. On the segmentation side, PIXray-based instance segmentation work [17] and occlusion-aware bilayer modeling that leverages PIDray-A/PIXray-A [35] both reinforce that improving masks under heavy overlap is a central requirement for end-to-end performance.

Evaluation, Generalization, and Deployment Considerations

Reported performance in X-ray security depends strongly on evaluation setup, since datasets differ in label type (image-level, boxes, masks), difficulty, and screening conditions. PIDray explicitly supports robustness testing through easy/hard/hidden splits [13], while PIXray and occlusion-extended variants (PIDray-A/PIXray-A) emphasize that segmentation quality can degrade sharply under overlap and may require additional supervision to evaluate occlusion behavior [17, 35]. Long-tail and rare-threat behavior is also a recurring concern, motivating both few-shot benchmarking [26] and debiasing-oriented methods for long-tailed detection [42]. For controlled evaluation under imbalance, curated balanced subsets have also been proposed [41].

Generalization is inconsistent across benchmarks: comparative studies across multiple public datasets report that no single model family is best everywhere, especially under dataset/scanner shift [32, 33]. This interacts with deployment constraints, where lightweight models aim to meet real-time requirements [23, 30, 40], yet comparability can be limited when high-speed approaches are evaluated primarily on private datasets [23, 37]. Complementary image-level threat recognition under severe imbalance has also been studied with transformer-based models, underscoring that task framing (classification vs. instance-level detection/segmentation) influences both evaluation and operational trade-offs [38].

Gap Summary and Motivation

Despite steady progress, prohibited-item detection and segmentation in X-ray security imagery remain constrained by overlap-driven feature mixing, weak texture/low contrast, and cluttered baggage layouts, which contribute to missed detections, false alarms, and unstable mask boundaries [14, 16, 17, 32]. Long-tailed categories and rare threats further challenge reliability, motivating dedicated benchmarks and debiasing strategies [13, 26, 42]. At the same time, generalization across datasets and scanners remains inconsistent, and some results are difficult to compare when they rely on private evaluation setups [23, 32, 33, 37].

Existing methods often improve one aspect of the pipeline at a time—for example, overlap-aware modules and anti-overlap transformer formulations [15, 18, 39, 43], material/feature cues [28, 29, 36], or occlusion-aware segmentation requiring extra supervision [35]. Motivated by these gaps, this proposal targets improving both detection and instance segmentation performance for prohibited items in X-ray imagery, with emphasis on robustness to overlap, clutter, and realistic difficulty settings [13, 17, 35].

Objective Statement

This research aims to improve the reliability and deployment readiness prohibited-item detection and segmentation in X-ray imagery by combining both a systematic literature review and an experimental study on a realistic benchmark. We aim to:

- Conduct a Systematic Literature Review (SLR) to map the research landscape (datasets, detection/segmentation methods, overlap handling, domain shift) and to extract recurring limitations and motivate a specific research target.
- Select and formalize a target failures from the SLR, including overlap-driven ambiguity, background confusion that cause false alarms and missed detections in cluttered baggage scenses.
- Establish reproducible baselines on PIDray (and its difficulty settings, including "hard" and "hidden" item cases) using an efficiency-oriented CNN detector (YOLO11s) and a real-time transformer detector (RT-DETRv2) models.
- Perform diagnostic error analysis (e.g. false positives and false negatives) to identify where failures concentrate and what factors correlate with missed detections and false alarms.

Research Questions

Prior studies show that X-ray screening is hardest when items are heavily overlapped, visually blend into the background, or appear rarely in training data. Building on these gaps, we focus on the following questions:

1. **RQ1**: How can X-ray detectors better separate true prohibited items from confusing background structures, so that false alarms and missed detections both decrease in cluttered baggage scans?
2. **RQ2**: Which training choices and overlap-aware modeling ideas most reduce the accuracy drop on deliberately concealed items (e.g., PIDray hard/hidden), and do these gains carry over to mask prediction when segmentation labels are available?
3. **RQ3**: What accuracy–efficiency trade-offs are achievable with lightweight CNN and transformer-based detectors, and which designs remain usable under real-time screening constraints?
4. **RQ4**: What failure modes dominate in practice (background confusion, poor localization, extreme overlap, or mask boundary errors), and how can targeted diagnostics guide the next round of model improvements?

Project Work Plan

Phase	Start Week	End Week	Task
Term 252 (Jan 11 – May 10, 2026)			
Phase 1	Week 6	Week 8	Data extraction & quality assessment
	Week 8	Week 10	Synthesis, gap analysis & SLR write-up
M1: SLR Completed (Week 10)			
Phase 2	Week 10	Week 12	Structure outline & write first draft
	Week 12	Week 13	Internal revisions with supervisor
M2: Draft Ready (Week 13)			
Summer Gap (May 21 – Aug 18, 2026): No project work scheduled			
Term 261 (Aug 19 – Dec 10, 2026)			
Phase 3	Week 1	Week 2	Finalize detection/segmentation protocol
	Week 2	Week 3	Define metrics, baselines & ablation plan
M3: Methodology Frozen (Week 3)			
Phase 4	Week 3	Week 4	Baseline training & initial evaluation
	Week 4	Week 5	Ablation studies & model refinement
	Week 5	Week 6	Error analysis (split-wise, per-class, boundary)
M4: Results Finalized (Week 6)			
Phase 5	Week 6	Week 8	Write analysis, results & discussion
	Week 8	Week 9	Prepare figures, tables & visual materials
	Week 9	Week 10	Compile & finalize complete paper
M5: Final Paper Complete (Week 10)			

Table 2: Project Work Plan.

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